The CPU and the fetch–execute cycle

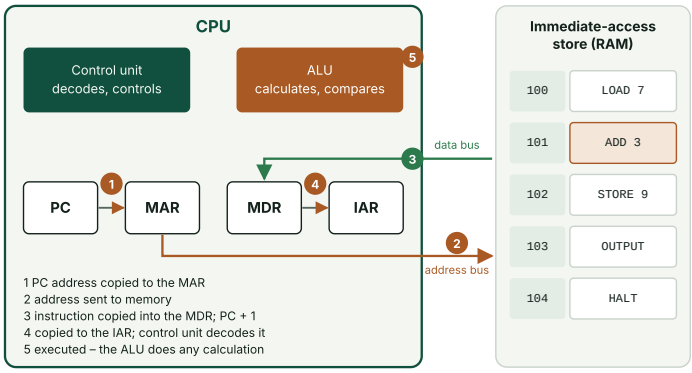
L13 FACT FILE

Specification: students should be able to

- Explain the purpose of the central processing unit (CPU)
- Describe the role of the following components of the CPU: the arithmetic logic unit (ALU), control unit and immediate-access store
- Describe the role the following play in the fetch-execute cycle: program counter, memory address register (MAR), memory data register (MDR), instruction address register (IAR) and ALU

CCEA fact file: Fact File 11: Computer Hardware, Computer Hardware 2

Key words: CPU, microprocessor, control unit, control signal, ALU, accumulator, immediate-access store, memory address, register, program counter, MAR, MDR, IAR, fetch, decode, execute, fetch–execute cycle



Inside the CPU: the fetch–execute cycle using the PC, MAR, MDR and IAR with main memory

What the CPU is for

The central processing unit (CPU) is the part of a computer that does the processing. It is a microprocessor, which means all of its circuits are built onto a single chip. Its purpose is to process data and carry out the instructions that make up computer programs. Every program, from a game on your phone to the till software in a shop in Belfast, is really a long list of simple instructions held in memory. The CPU works through these instructions one after another by repeatedly fetching an instruction from memory, decoding it to find out what it means, and then executing it. This repeating process is called the fetch–execute cycle. While doing this, the CPU also manages the rest of the computer by sending control signals to memory and to input and output devices. People often call the CPU the brain of the computer, but in an exam you must explain what it actually does: it processes data and executes program instructions.

Control unit, ALU and immediate-access store

The CPU has three main components. The control unit manages the running of instructions. It decides which instruction is next, fetches it, decodes it and then sends control signals to the other parts so that the instruction is carried out. It co-ordinates everything but does not do the calculations itself. The arithmetic logic unit (ALU) does the calculations and the logic. Arithmetic means adding, subtracting, multiplying and dividing. Logic means comparisons and the operators AND, OR and NOT, which let a program make decisions, such as checking whether a pupil has enough money on their canteen card. The ALU contains a special register called the accumulator, which holds the result of each calculation until it is stored back in memory. The immediate-access store (IAS) is the main memory. It holds the programs and data currently in use. Each memory location has a unique address, so the CPU can go straight to the instruction or data it needs.

Registers and what they hold

A register is a very small, very fast memory location inside the CPU, used for one particular job. Five matter at GCSE. The program counter (PC) holds the address of the next instruction to be fetched. Because most instructions run in order, it goes up by one after each fetch. The memory address register (MAR) holds the address of the memory location that is about to be read from or written to. The memory data register (MDR) holds the actual instruction or data that has just been fetched from memory, or that is about to be written to memory; all data going between the CPU and main memory passes through it. The instruction address register (IAR), as CCEA describes it, holds the instruction currently being decoded and executed; some textbooks call this the instruction register. Finally, the accumulator in the ALU holds the result of the latest calculation. A quick way to remember the difference: the MAR holds where, the MDR holds what.

The cycle step by step

In the fetch stage, the address held in the program counter is copied into the MAR. The control unit then reads that memory location, and the instruction stored there is copied into the MDR and then into the IAR. The program counter is increased by one so that it points to the next instruction. In the decode stage, the control unit works out what the instruction means, for example whether it is asking for a value to be loaded, two numbers to be added or a result to be stored. In the execute stage, the instruction is carried out. If it needs data from memory, the data's address goes into the MAR and the data comes back through the MDR. If it needs a calculation or comparison, the ALU does it and the result goes into the accumulator. The cycle then starts again with the next address in the program counter. A modern CPU repeats this cycle billions of times every second.

Key terms from the mark schemes

CPU
Central processing unit. A microprocessor whose purpose is to process data and carry out the instructions of programs, managing the functions of the computer.
Credited: Manages the functions of a computer by processing data and instructions

ALU
Arithmetic logic unit: the part of the CPU that carries out arithmetic calculations and logical operations and comparisons.
Credited: Arithmetic and logic unit

Control unit
The part of the CPU that fetches and decodes instructions and sends control signals to co-ordinate the other components.
Credited: Directs/co-ordinates the operations of the CPU

Immediate-access store
Main memory (RAM) holding the programs and data currently in use; each location has a unique address.
Credited: IAS; main memory

Register
A very small, very fast memory location inside the CPU used for a specific purpose.

Program counter
Register holding the address of the next instruction to be fetched; incremented after each fetch.
Credited: PC

MAR
Memory address register: holds the address of the memory location being read from or written to.

MDR
Memory data register: holds the data or instruction being transferred to or from memory.
Credited: Memory buffer register

IAR
Instruction address register (CCEA term): holds the instruction currently being decoded and executed.
Credited: Instruction register (IR)

Accumulator
Register in the ALU that holds the result of a calculation before it is stored in memory.
Credited: ACC

Fetch–execute cycle
The repeated process in which the CPU fetches an instruction from memory, decodes it and executes it.
Credited: Fetch–decode–execute cycle

Exam focus

This topic comes up nearly every year. You have been asked to explain the purpose of the CPU (2023 Q5, 2024 Q5), describe the ALU and control unit (2019 Q5(c)–(d)), explain registers and the cycle (2021 Q5, 2025 Q7(a)–(b)), and write a levelled answer on the ALU and immediate-access store (2022 Q4). Marks are given for each correct role, so 'describe' means say what the part does, not just name it. For the CPU's purpose, say it processes data and carries out program instructions; 'it is the brain' scores nothing on its own. Write every acronym in full at least once, as examiners regularly report lost marks for this. In fetch–execute questions, give the steps in the right order and name the register used at each step. Common mistakes are saying the ALU decodes instructions, mixing up the MAR and the MDR, and forgetting that the program counter is incremented. In levelled questions, use correct technical terms and link each point to the cycle.

Unit 1 2025 Q7	[8]	Fetch–execute cycle; input devices and uses
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2023 Q5	[10]	CPU, system software resources, real-time vs multi-user, utilities, I/O devices
Unit 1 2022 Q4	[4]	ALU and immediate-access store (levelled)
Unit 1 2021 Q5	[6]	CPU, registers, speed, clock speed

? Check yourself

1. Explain the purpose of the central processing unit (CPU). [2]
2. State the three main components of the CPU. [3]
3. Describe the role of the arithmetic logic unit (ALU). Give an example of an operation it carries out. [3]

Answers: 1. It processes data (1) / it carries out (executes) the instructions of programs (1). Also accept: manages/controls the functions of the computer by sending control signals (1). 'Brain of the computer' alone = 0. 2. Control unit (1) / arithmetic logic unit (ALU) (1) / immediate-access store (IAS) or main memory (1). 3. Carries out arithmetic calculations (1) / carries out logical operations or comparisons, e.g. AND, OR, NOT, greater than (1) / valid example, e.g. adding two numbers or checking if a balance is below a limit (1). Accept: results held in the accumulator (1).